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EXPERIMENT 10

Performance optimization

Aim

Measure and improve page, report and workflow performance with selective queries and bounded processing.

Procedure

1. Create a synthetic dataset within plan limits (e.g. 200 requests, 20 technicians, 8 bays); record actual sizes.

2. Measure the command center page, Active_Requests report and search repeatedly: one warm-up, then at least ten timings; report median and range.

3. Replace full-table loops with criteria-based fetches; call APIs outside loops.

4. Limit initial report rows and columns; load heavy content only where needed (Board shows active statuses only).

5. Change one thing at a time and re-measure; confirm totals and permissions are unchanged. Do not claim access to arbitrary database indexes.

```
// Select only needed records
active = Service_Request[Status != "Delivered"];
readyCount = Service_Request[Status == "Ready for Pickup"].count();
```

Implementation and verification

Operation	Before median	After median	Correctness
Command center page load	___ ms	___ ms	Same panels
Active_Requests report	___ ms	___ ms	Same IDs and count
Owner search	___ ms	___ ms	Same results

Evidence and testing

E10: dataset size, timing table with median and range, notes on each change. Leave blank until you actually measure.

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Application images and explanation

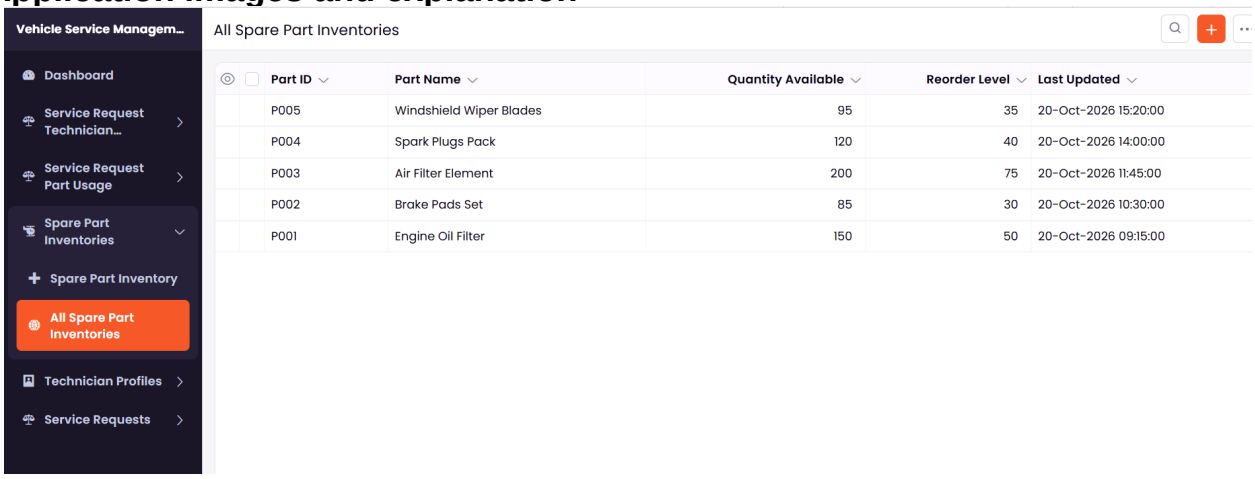
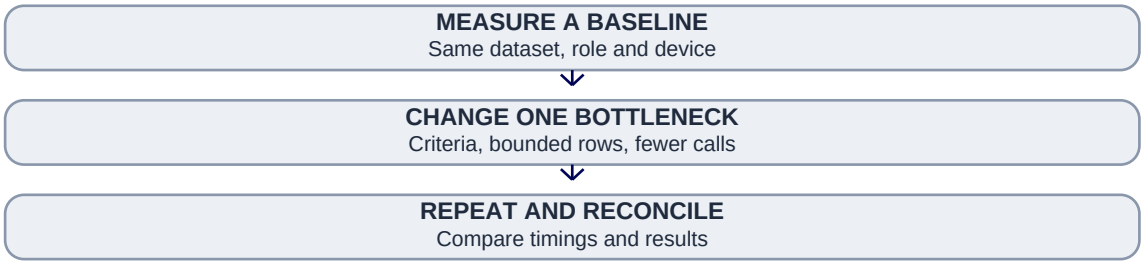


Figure 1: All Spare Part Inventories report used as a performance test surface

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Result

Expected result and verification criteria. Improvements are reported with test conditions and unchanged results; no guarantee is inferred from a small dataset.